

UNIT-4 | Lecture 2

Armstrong's Axioms & Inference Rules

Coverage: Hour 2

Learning Objectives

Understand Armstrong's three basic axioms.

Apply derived inference rules to functional dependencies.

Basic Axioms

Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$.

Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$.

Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$.

Derived Rules

Union: $X \rightarrow Y$ and $X \rightarrow Z \Rightarrow X \rightarrow YZ$.

Decomposition: $X \rightarrow YZ \Rightarrow X \rightarrow Y$ and $X \rightarrow Z$.

Pseudotransitivity: $X \rightarrow Y$ and $WY \rightarrow Z \Rightarrow WX \rightarrow Z$.

Examples

Given $A \rightarrow B$ and $B \rightarrow C$, transitivity gives **$A \rightarrow C$** .

Given $A \rightarrow B$ and $A \rightarrow C$, union gives **$A \rightarrow BC$** .

Given $A \rightarrow BC$, decomposition gives **$A \rightarrow B$** and **$A \rightarrow C$** .

Importance

Armstrong's axioms are **sound and complete** for functional-dependency inference. They help derive implied dependencies, find attribute closure, and reason about candidate keys and normalization.

Key Takeaway

Armstrong's axioms provide a systematic method for deriving all logically implied functional dependencies.